

HAS-RL: Human-Adaptive Safe Reinforcement Learning for Clinician-Configured Force Safety in Robotic Ultrasound Spinal Scanning

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Abstract

Robotic ultrasound (US) imaging promises to standardize diagnostic scanning and reduce sonographer fatigue, but any assistive robotic system in physical contact with a patient must respect a hard force-safety limit that is patient-, tissue-, and operator-dependent. Prior safe reinforcement learning (RL) for robotic US trains a policy against a *single, fixed* force threshold, so a change in the tolerated force requires retraining; recent bandit-based safe-RL multiplier methods, in turn, were designed for the *offline* setting and inject destabilizing non-stationarity when transplanted into an online policy-gradient loop. We present **HAS-RL**, a framework with four coupled contributions. (i) *Operator-conditioned zero-shot safety*: instead of a fixed maximum force, we place the operator’s runtime safety setting into the observation and train one policy over a randomized curriculum. The resulting policy adapts its contact force to that setting at deployment with no retraining, tracking a mid-scan dial change without breaching the active limit. (ii) The *Smoothed Bandit-Lagrangian* (SBL), which drives an EXP3 multiplier bandit but applies an exponential-moving-average (EMA) of the multiplier *expectation* to the policy update. Over three seeds SBL attains the highest mean image quality (0.603) and the most stable training (lowest per-epoch violation standard deviation, and the tightest multiplier trajectory, $\lambda_{\text{std}} = 0.42 \pm 0.02$) among the multiplier strategies while matching the perfect constraint satisfaction of standard dual ascent. (iii) *Anticipatory tracking* via a short look-ahead of the future lateral error, which reduces root-mean-square lateral tracking error by 28% ($0.084 \rightarrow 0.061$). (iv) *Two-sided acoustic-window safety*: we model ultrasound force safety not as a one-sided harm ceiling but as a clinician-specified window $[\kappa_{\text{low}}, \kappa_{\text{high}}]$. A single conditioned policy keeps the contact force inside any specified window (zero out-of-window violations, 100% residence) and now *uses* the range a ceiling-only model wasted—reaching 10.2N in a $[9, 12]$ N window versus 7.97N under a bare $\kappa = 12$ N ceiling. We release a fully synthetic US environment (kinematics, low-pass impedance force control, and a B-mode speckle/bone/shadow image model) so that all results are reproducible on a CPU.

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1 Introduction

The objective is automated assistance for diagnostic medical ultrasound, with emphasis on patient-contact safety and clinician-configurable operation. Ultrasound is inexpensive, portable, and radiation-free, but its diagnostic value is critically dependent on the mechanical skill of the operator: the probe must be kept perpendicular to and centered on the target anatomy, and it must be pressed with enough force to establish acoustic coupling through the skin, yet not so much force as to harm the patient. Assisting this skill with a robot is attractive for both throughput and standardization, but the physical human–robot contact makes safety a first-class, non-negotiable requirement rather than a soft objective.

Reinforcement learning (RL) is a natural fit for learning the force/pose policy because the coupling between applied force and image quality is nonlinear and hard to model in closed form. However, a reward-maximizing agent will happily exploit the fact that *more force yields better acoustic coupling and therefore higher image quality*, pressing arbitrarily hard unless it is explicitly constrained. This is the core tension that safe RL must resolve for medical robotics.

Two recent lines of work frame the problem but each leaves a gap that this paper targets:

- **Constraint-aware policy optimization for robotic US [2].** As the pioneering application of formal safe RL to this domain, this work formulates US force regulation as a constrained Markov decision process (CMDP). It solves this by embedding trust-region policy updates inside a primal–dual interior-point solver, augmented with an imitation-based learning rate schedule to stabilize consecutive policy shifts. While it is the foundational precedent for safe robotic US, the force constraint is a *single fixed threshold* baked into

training. A different tolerated force (a frail patient, a different body region, a cautious clinician) requires retraining a different policy.

- **Online optimization for offline safe RL (O3SRL)** [3]. This work uses an EXP3 multi-armed bandit to select the Lagrange multiplier online *during offline training*, obtaining strong regret guarantees. The multiplier is context-free (a single budget) and, as we show empirically (Sec. 7.4), sampling a discrete multiplier arm every epoch and applying it directly to an *online* policy-gradient advantage injects large variance into the actor’s learning signal.

Contributions. We introduce **HAS-RL (Human-Adaptive Safe RL)** and make four contributions, validated with controlled, single-variable ablations (same seeds, same budget):

1. **Operator-conditioned zero-shot safety (dynamic κ).** Instead of a fixed $F_{\max}$, we place the operator’s runtime safety setting into the observation and train one policy over a randomized curriculum. The resulting policy *adapts its contact force to that setting at deployment with no retraining*, tracking a mid-scan dial change and never breaching the active limit (Sec. 7.2).
2. **Smoothed Bandit-Lagrangian (SBL).** We keep the appeal of a bandit over discrete multiplier “arms” introduced by O3SRL [3] (which provides a bounded, interpretable penalty range). However, to successfully port this offline method to an online setting, we apply an EMA of the multiplier *expectation* to the policy update. This removes the severe per-epoch gradient jumps that otherwise destabilize a direct online adaptation of the O3SRL bandit (Sec. 5.2).
3. **Anticipatory tracking.** We augment the state with a short look-ahead of the future lateral error, coupling trajectory anticipation with force safety and measurably tightening tracking on curved (scoliotic) anatomy (Sec. 7.3).
4. **Two-sided acoustic-window safety.** We model ultrasound force safety not as a one-sided harm ceiling but as a clinician-specified *window* $[\kappa_{\text{low}}, \kappa_{\text{high}}]$ —a minimum diagnostic contact force and a maximum safe force—matching the physical acoustic window. The lower bound also removes a failure mode of ceiling-only safety: it closes the *headroom gap* (the policy uses the force range a ceiling leaves unused) and makes the safety constraint genuinely *binding* rather than shadowed by the reward (Sec. 7.5).

We also contribute the open, dependency-light simulator used throughout, which produces synthetic B-mode frames and runs entirely on CPU.

Honest scope. We are deliberately explicit about what each contribution buys. The zero-shot κ result is strong and, to our knowledge, novel in this domain. The SBL result is favorable but modest: on our reward design SBL edges out standard dual ascent on image quality, and its clearest, cleanest advantage is *training stability* relative to the unsmoothed bandit; the quality gap to dual ascent lies within one seed standard deviation, and we say so (Sec. 7.4, Sec. 8).

2 Related Work

Safe RL and constrained MDPs. The standard formalism for safety-constrained control is the constrained MDP (CMDP) [4], in which one maximizes expected return subject to an expected-cost budget. Two broad solution families exist: *trust-region* / *projection* methods such as Constrained Policy Optimization (CPO) [5], which enforce the constraint through a per-step trust region, and *Lagrangian* methods, which relax the constraint into a penalty with a multiplier

λ optimized by dual ascent [6, 7]. Lagrangian PPO is the workhorse of this paper’s baseline because it is simple, scalable, and requires no second-order information.

Stabilizing the Lagrange multiplier. A well-known pathology of dual ascent is oscillation: the multiplier over- and under-shoots, causing the policy to alternate between reckless and over-conservative behavior. Stooke et al. [8] address this with a PID controller on the constraint error (“PID-Lagrangian”), adding derivative and integral terms to damp oscillation. SBL is a close cousin in *intent*—smoothing the multiplier signal—but differs in mechanism: rather than a PID controller on a continuous multiplier, SBL maintains a bandit distribution over a bounded, discrete set of multiplier arms and low-pass filters its *expectation*. The bandit provides a hard, interpretable ceiling on the penalty; the EMA provides the smoothing.

Bandit multipliers and offline safe RL. O3SRL [3] casts the online selection of the dual variable as an adversarial bandit (EXP3 [9]) and targets the *offline* safe-RL setting. Our SBL borrows the EXP3 machinery but is built for *online* policy gradients, where we find that applying a freshly sampled discrete multiplier every epoch is destabilizing; SBL’s smoothing is precisely the fix.

Robotic ultrasound navigation and safe RL. Robotic-assisted ultrasound navigation via reinforcement learning was demonstrated by Hase et al. [1] for spinal/sacrum examination. Building on this, Duan et al. [2] pioneered the formal application of Safe RL to this domain. They formulate US force regulation as a CMDP to manage the complex coupling between the applied force profile and ultrasound image quality. Technically, they solve this using a Constrained Policy Optimization (CPO) approach where gradient-based policy updates are computed via a primal–dual interior-point technique to strictly bound the force magnitude. Furthermore, to ensure stable learning, they explicitly design the policy gradient learning rate from an imitation perspective, forcing consecutive policies to remain close. While their method mathematically guarantees monotonic improvement within a fixed safety envelope, the constraint remains a *single fixed threshold* baked into the training process. HAS-RL’s central departure is to make that threshold a *conditioning input*, yielding a continuum of safety behaviors from a single policy—something neither a fixed-threshold domain method nor a single-budget offline method produces.

Foundational policy-gradient tools. Throughout we use Proximal Policy Optimization (PPO) [10] with clipped surrogate objectives and Generalized Advantage Estimation (GAE) [11] for both the reward and the cost critic.

3 Preliminaries

3.1 Constrained Markov decision process

We model the scan as a CMDP $(\mathcal{S}, \mathcal{A}, P, r, c, \gamma)$ with states $s \in \mathcal{S}$, actions $a \in \mathcal{A}$, transition kernel P , reward r , a per-step safety cost $c \in \{0, 1\}$, and discount γ . For a policy π_θ we define the return $J_r(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$ and the discounted cost $J_c(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t c_t]$. The constrained objective is

$$\max_{\theta} J_r(\pi_\theta) \quad \text{s.t.} \quad \mathbb{E}[c] \leq d, \quad (1)$$

where d is the tolerated per-step violation rate (we use $d = 0.02$).

3.2 PPO with twin critics and GAE

We optimize π_θ using the Proximal Policy Optimization (PPO) [10] clipped surrogate objective:

$$\mathcal{L}^{\text{clip}}(\theta) = \mathbb{E}_t \left[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad \rho_t = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}, \quad (2)$$

with $\epsilon = 0.2$. Because the task reward and safety cost are distinct, and often conflicting, signals, we maintain *twin* critics: a reward critic $V_R(s)$ and a cost critic $V_C(s)$. To elegantly trade off variance and bias in our advantage estimates, we employ Generalized Advantage Estimation (GAE) [11, 12]. Rather than an infinite-horizon sum, GAE is computed over finite-length rollout trajectories of length T , explicitly handling episode boundaries. For the reward signal, we first compute the 1-step temporal-difference (TD) error at each step t :

$$\delta_t^R = r_t + \gamma(1 - d_t)V_R(s_{t+1}) - V_R(s_t), \quad (3)$$

where $d_t \in \{0, 1\}$ indicates whether the episode terminated at step t . The reward GAE is then the exponentially discounted sum of these TD errors up to the end of the trajectory:

$$\hat{A}_t^R = \sum_{l=0}^{T-t-1} (\gamma \lambda_{\text{GAE}})^l \delta_{t+l}^R, \quad (4)$$

where $\gamma = 0.99$ is the discount factor and $\lambda_{\text{GAE}} = 0.95$ is the trace-decay parameter. This is computed efficiently via the backward recursive relation $\hat{A}_t^R = \delta_t^R + \gamma \lambda_{\text{GAE}}(1 - d_t)\hat{A}_{t+1}^R$. The cost advantage $\hat{A}_t^C$ is derived identically using the cost signal c_t and the cost critic $V_C(s)$. Both advantages are normalized independently prior to policy updates.

3.3 Lagrangian relaxation

The Lagrangian of the CMDP in (1) is

$$\mathcal{L}(\theta, \lambda) = J_r(\pi_\theta) - \lambda(J_c(\pi_\theta) - d), \quad \lambda \geq 0. \quad (5)$$

In our online policy-gradient framework, this relaxation dictates that the actor is updated using a composite *safe advantage*:

$$\hat{A}_t^{\text{safe}} = \hat{A}_t^R - \lambda \hat{A}_t^C. \quad (6)$$

The Lagrangian multiplier λ governs the dynamic tradeoff between seeking rewards and penalizing constraint violations. How this λ is set and stabilized at each epoch is the central algorithmic design axis studied in Sec. 5.2 and ablated in Sec. 7.4.

4 The HAS-RL Ultrasound Simulator

Building upon the spinal examination and navigation paradigm established by Hase et al. [1], we note a critical limitation for Safe RL in their open-source environment: it relies on navigating an offline grid-world of pre-recorded, static volunteer images. Because the images are pre-recorded, the applied contact force is fixed at the time of data collection, making it impossible for an agent to dynamically alter pressure or experience real-time impedance physics. To study continuous force safety, all our experiments use a custom-built, self-contained dynamic physics simulator (no external assets, CPU-only). It captures the three physical ingredients relevant to real-time safety: probe kinematics, low-pass impedance force control, and a synthetic B-mode image model that dynamically renders based on instantaneous pressure. An episode is a single head-to-waist sweep of $T = 100$ steps.

4.1 Kinematics and control

The probe advances at constant longitudinal velocity $v_x = 1/T$ along a normalized axis $x \in [0, 1]$. The ground-truth spinal centerline is a scoliotic sinusoid

$$y^*(x) = A \sin(2.5\pi x), \quad A = 0.07, \quad (7)$$

which the probe must laterally track. The action $a = (a_{\text{lat}}, a_{\text{force}}) \in [-1, 1]^2$ controls lateral steering and the force *rate*:

$$y \leftarrow \text{clip}(y + 0.022 a_{\text{lat}}, -0.25, 0.25), \quad (8)$$

$$F \leftarrow \text{clip}(F + 0.7 a_{\text{force}}, 0, 20) \text{ [N]}. \quad (9)$$

The force update is a low-pass impedance model: the agent commands a bounded *change* in force (max 0.7 N/step), not an instantaneous force, which mirrors the compliant behavior of a real force-controlled end effector. The lateral error is $e(x) = y - y^*(x)$.

4.2 Synthetic B-mode image model

Image quality $q \in [0, 1]$ is a differentiable proxy that rewards two things a sonographer cares about—adequate acoustic coupling and lateral centering:

$$q(F, e) = \begin{cases} 0.05 (F/F_{\min}), & F < F_{\min}, \\ 0.65 \underbrace{e^{-\frac{1}{2} \left(\frac{F-F_{\text{opt}}}{3.2} \right)^2}}_{\text{acoustic coupling}} + 0.35 \underbrace{e^{-\frac{1}{2} \left(\frac{e}{0.015} \right)^2}}_{\text{centering}}, & F \geq F_{\min}, \end{cases} \quad (10)$$

with $F_{\min} = 1.0 \text{ N}$ (below which gel/tissue coupling is lost) and $F_{\text{opt}} = 7.0 \text{ N}$ (peak acoustic window). Crucially, quality *increases with force* up to F_{opt} , which is exactly why an unconstrained agent presses too hard. For qualitative inspection the simulator can also render a full 2D B-mode frame (Fig. 2) with Rayleigh speckle, time-gain-compensated depth attenuation, a hyperechoic parabolic bone arc whose lateral position follows e , and an acoustic shadow beneath the bone.

4.3 Reward and cost

The composite reward densely shapes tracking, quality, and control effort:

$$r = \underbrace{5.0 q}_{\text{quality}} + \underbrace{3.0 e^{-\frac{1}{2} (e/0.015)^2}}_{\text{centering bonus}} - \underbrace{35.0 e^2 - 4.0 |e|}_{\text{tracking penalty}} - \underbrace{0.5 (F - F_{\text{tgt}})^2}_{\text{budget-aware force}} - \underbrace{0.15 \|a - a_{\text{prev}}\|^2}_{\text{smoothness}} - \underbrace{0.05 \|a\|^2}_{\text{effort}}, \quad (11)$$

where the budget-aware target $F_{\text{tgt}} = \min(F_{\text{opt}}, \kappa - 0.4)$ pulls the force toward the highest *safe* value for the current ceiling. The per-step safety cost is a hard indicator of exceeding the operator ceiling:

$$c = \mathbf{1}[F > \kappa]. \quad (12)$$

We note explicitly (and return to it in Sec. 8) that the budget-aware term makes the reward itself partly safety-aware; this is realistic (a good controller should not aim for the ceiling) but it means the Lagrangian mechanism is not the sole source of safety.

4.4 Observation: exposing the ceiling and the future

The 10-dimensional observation is

$$s = [x, y, e, \dot{e}, F, \dot{F}, q, \boxed{\kappa}, \boxed{e^{+5}}, \boxed{e^{+15}}], \quad (13)$$

where $\dot{e}, \dot{F}$ are one-step differences. The three boxed entries carry the paper’s architectural novelties: κ is the **operator ceiling** (Sec. 5.1), and $e^{+5} = y - y^*(x + 5v_x)$, $e^{+15} = y - y^*(x + 15v_x)$ are the **look-ahead** lateral errors 5 and 15 steps into the future (Sec. 5.3). With look-ahead disabled (an ablation), these two entries are set to zero and the network architecture is otherwise unchanged.

5 Method

5.1 Operator-conditioned zero-shot safety (dynamic κ)

Rather than train one policy per force limit, we make the limit an input. Because κ appears in the observation (13), in the budget-aware reward target $F_{\text{tgt}} = \min(F_{\text{opt}}, \kappa - 0.4)$, and in the cost (12), the policy $\pi_\theta(a \mid s, \kappa)$ and both critics $V_R(s, \kappa), V_C(s, \kappa)$ learn a *continuum* of behaviors. During training we sample a fresh ceiling $\kappa \sim \mathcal{U}[3, 12]$ N at the start of every episode (a curriculum over constraints). At deployment the clinician sets—or turns— κ freely within the trained range, and the single policy responds *zero-shot*: no gradient steps, no new data. This is the capability neither a fixed-threshold domain method [2] nor a single-budget offline method [3] provides.

5.2 Smoothed Bandit-Lagrangian (SBL)

SBL sets the multiplier λ in (5) each epoch. We keep $K = 5$ discrete multiplier arms $\lambda_1, \dots, \lambda_5$ evenly spaced in $[0, 5]$ and an EXP3 weight vector $w \in \mathbb{R}_{>0}^K$ inducing the sampling distribution $p_i = w_i / \sum_j w_j$.

Bandit weight update. After the policy update, we score the sampled arm by its realized Lagrangian value $\ell = \bar{r} - \lambda_{i_t}(\bar{c} - d)$ (mean batch reward $\bar{r}$, mean batch cost $\bar{c}$), normalize $(\ell/10)$ for numerical stability, form the standard importance-weighted estimate $\hat{\ell}_i = \ell \mathbf{1}[i = i_t]/p_i$ (clipped to $[-100, 100]$), and apply the exponential-weights update $w_i \leftarrow w_i \exp(-\eta \hat{\ell}_i)$ with $\eta = 0.05$, followed by renormalization. This is the O3SRL machinery [3]; SBL’s departure is entirely in how the multiplier is *consumed* by the policy to make it usable online. Two ideas define SBL:

(1) **Use the expectation, not a sample.** A naive online bandit samples an arm λ_{i_t} and applies it directly. That single-sample estimate has high variance, and because it multiplies the *advantage* in the policy update, its jumps propagate into the actor’s gradient. Instead we compute the exact expectation $\bar{\lambda}_t^{\text{exp}} = \sum_i p_i \lambda_i$, which carries no sampling variance.

(2) **Low-pass filter it.** We then apply an EMA with smoothing $\tau = 0.05$:

$$\lambda_t = (1 - \tau) \lambda_{t-1} + \tau \bar{\lambda}_t^{\text{exp}}, \quad (14)$$

so the multiplier that reaches the policy evolves smoothly across epochs. The active λ_t is used to form the safe advantage $\hat{A}^{\text{safe}} = \hat{A}^R - \lambda_t \hat{A}^C$. Algorithm 1 summarizes one epoch.

Algorithm 1 HAS-RL: PPO with GAE and Smoothed Bandit-Lagrangian

```
1: Initialize actor  $\theta$ , critics  $\phi_R, \phi_C$ , state  $s_1$ , arms  $\{\lambda_i\}_{i=1}^N$ , weights  $w_i = 1$  for  $i = 1 \dots N$ 
2: for iteration  $k = 1, 2, \dots$  do
3:    $\theta_{\text{old}} \leftarrow \theta$ 
4:    $(s_1, \kappa_1, a_1, r_1, c_1, \dots, s_T) = \text{rollout}(s_1, \pi_{\theta_{\text{old}}})$   $\triangleright$  Conditioned on safety window  $\kappa$ 
5:    $v_t^R = V_{\phi_R}(s_t, \kappa_t)$  and  $v_t^C = V_{\phi_C}(s_t, \kappa_t)$  for  $t = 1 \dots T$ 
6:    $(A_{1:T}^R, y_{1:T}^R) = \text{GAE}(r_{1:T}, v_{1:T}^R, \gamma, \lambda_{\text{GAE}})$   $\triangleright$  Reward advantage and returns
7:    $(A_{1:T}^C, y_{1:T}^C) = \text{GAE}(c_{1:T}, v_{1:T}^C, \gamma, \lambda_{\text{GAE}})$   $\triangleright$  Cost advantage and returns
8:    $p_i = \frac{w_i}{\sum_{j=1}^N w_j}$  for  $i = 1 \dots N$ ; sample arm  $i_k \sim p$ 
9:    $\bar{r} = \frac{1}{T} \sum_{t=1}^T r_t$ ;  $\bar{c} = \frac{1}{T} \sum_{t=1}^T c_t$ 
10:   $\lambda_k \leftarrow (1 - \tau)\lambda_{k-1} + \tau \sum_{i=1}^N p_i \lambda_i$   $\triangleright$  SBL multiplier (Eq. (14))
11:   $A_t^{\text{safe}} = A_t^R - \lambda_k A_t^C$  for  $t = 1 \dots T$   $\triangleright$  Composite safe advantage
12:  for  $m = 1 \dots M$  do
13:     $\rho_t = \frac{\pi_{\theta}(a_t|s_t, \kappa_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t, \kappa_t)}$  for  $t = 1 \dots T$ 
14:     $\tilde{\rho}_t = \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon)$  for  $t = 1 \dots T$ 
15:     $\mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^T [\min(\rho_t A_t^{\text{safe}}, \tilde{\rho}_t A_t^{\text{safe}}) + c_{\text{ent}} \mathbb{H}(\pi_{\theta}(\cdot|s_t, \kappa_t))]$ 
16:     $\mathcal{L}(\phi_R) = \frac{1}{T} \sum_{t=1}^T (V_{\phi_R}(s_t, \kappa_t) - y_t^R)^2$ 
17:     $\mathcal{L}(\phi_C) = \frac{1}{T} \sum_{t=1}^T (V_{\phi_C}(s_t, \kappa_t) - y_t^C)^2$ 
18:    Update  $\theta, \phi_R, \phi_C$  via gradient descent on their respective losses
19:  end for
20:   $\ell \leftarrow \bar{r} - \lambda_{i_k}(\bar{c} - d)$   $\triangleright$  Compute realized Lagrangian for sampled arm
21:   $\hat{\ell}_i \leftarrow \text{clip}(\ell/10, -100, 100) \frac{\mathbb{1}_{[i=i_k]}}{p_i}$  for  $i = 1 \dots N$   $\triangleright$  Importance-weighted loss
22:   $w_i \leftarrow w_i \exp(-\eta \hat{\ell}_i)$  for  $i = 1 \dots N$   $\triangleright$  EXP3 weight update
23: end for
```

Ablation modes. To isolate SBL’s effect we implement two alternative multiplier rules behind a single flag, holding everything else fixed:

- **dual**: projected dual ascent, $\lambda_t = \text{clip}(\lambda_{t-1} + \eta_{\lambda}(\bar{c} - d), 0, \lambda_{\text{max}})$, $\eta_{\lambda} = 0.5$ (standard PPO-Lagrangian).
- **exp3_raw**: use the sampled arm directly, $\lambda_t = \lambda_{i_t}$ (the O3SRL-style online bandit, no smoothing).
- **sbl**: expectation + EMA, Eq. (14) (ours).

5.3 Anticipatory tracking

A reactive controller that only sees the current lateral error lags a curved target, because by the time the error is measured the probe has already advanced. We give the policy a short preview by appending the future lateral errors e^{+5}, e^{+15} to the state (13). This is a minimal, model-based form of preview control: the spine geometry (7) is a property of the patient anatomy the robot is scanning, so a short look-ahead is available in practice from a pre-scan or an atlas. We quantify its effect in Sec. 7.3.

5.4 From a single ceiling to an operator window (two-sided conditioning)

Conditioning on a single ceiling has a subtle failure mode: the policy has no incentive to *use* the headroom below the ceiling. At $\kappa = 12$ the learned force plateaus at 7.97 N (Table 2) because the budget-aware target $F_{\text{tgt}} = \min(F_{\text{opt}}, \kappa - 0.4)$ saturates at the acoustic optimum $F_{\text{opt}} = 7$ N; roughly 3.5 N of the operator’s range goes unused, so the high end of the dial is effectively

wasted. This motivates generalizing the scalar ceiling to an *operator window* $[\kappa_{\text{low}}, \kappa_{\text{high}}]$, in which the clinician specifies both the minimum diagnostic contact force and the maximum safe force. Three changes suffice, all behind a `two_sided` flag (the one-sided model above is unchanged): (i) the observation (13) gains κ_{low} ($10 \rightarrow 11$ dims); (ii) the cost becomes *out-of-window*, $c = \mathbf{1}[F < \kappa_{\text{low}} \text{ or } F > \kappa_{\text{high}}]$; and (iii) the reward target aims for the in-window force nearest the acoustic optimum, $F_{\text{tgt}} = \text{clip}(F_{\text{opt}}, \kappa_{\text{low}} + 0.4, \kappa_{\text{high}} - 0.4)$. The curriculum samples $\kappa_{\text{high}} \sim U[4, 12]$ and a width $w \sim U[1.5, 3]$ with $\kappa_{\text{low}} = \kappa_{\text{high}} - w$, so the window—and hence the force floor the policy must respect—slides up with the setting. We evaluate this model in Sec. 7.5. A one-sided ceiling is the degenerate case $\kappa_{\text{low}} \rightarrow 0$; the two-sided window is what makes the safety constraint bind.

6 Experimental Setup

Unless noted, agents use PPO ($\epsilon = 0.2$, 10 inner epochs, minibatch 64, actor lr 3×10^{-4} , critic lr 10^{-3} , gradient-norm clip 0.5), $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, and the network is a two-hidden-layer (128×128) tanh MLP for the actor mean and each critic; the actor log-std is a learned parameter clamped to $[-2, 0.5]$. The safety budget is $d = 0.02$. Ablations use three seeds $\{0, 1, 2\}$, 40,000 timesteps, and 2,000 steps/epoch, flipping exactly one variable per comparison. The headline zero-shot model is trained for 80,000 timesteps over the $\kappa \in [3, 12]$ curriculum. All runs are CPU-only and torch- and numpy-seeded.

We report: mean/max contact force, per-step *violations* (count of steps with $F > \kappa$), mean image quality q , root-mean-square (RMS) and max lateral tracking error, and two stability numbers—the standard deviation of per-epoch violations (`train_viol_std`) and of the active multiplier (`lambda_std`) across training.

Clinical Disclaimer. All force thresholds and force-window values used in this study are simulation parameters for algorithmic evaluation and should not be interpreted as clinically validated patient-safety thresholds.

7 Results

7.1 HAS-RL vs. unconstrained baseline

Table 1 contrasts both formulations of our framework—**HAS-RL (one-sided)** and the bounded window extension **HAS-RL (two-sided)**—against an unconstrained Vanilla PPO baseline that optimizes image quality alone, evaluated at an upper operator safety ceiling of $\kappa = 5$ N (with $[\kappa_{\text{low}}, \kappa_{\text{high}}] = [3, 5]$ N for the two-sided agent). The baseline maximizes acoustic coupling by pressing well past the ceiling—mean 5.25 N, and **96 of 100** steps in violation—to reach quality 0.90.

In contrast, both HAS-RL variants maintain strict safety with **zero violations**. HAS-RL (one-sided) holds mean contact force at 4.32 N with a maximum of 4.37 N (comfortably under 5 N) while achieving quality 0.80, accurately following the scoliotic spine curvature without lateral drift. HAS-RL (two-sided) regulates contact force within the $[3, 5]$ N window at mean 4.04 N (max 4.09 N) with quality 0.77. This demonstrates the intended, honest safety–quality tradeoff: the constrained agents forgo the excessive reward available only in the unsafe force regime while tracking the spinal curvature with high lateral fidelity. Figure 1 shows the full head-to-head trajectory—force vs. ceiling, cumulative violations, image quality, and lateral tracking of the spine centerline—for HAS-RL (one-sided) on a single shared environment rollout.

Table 1: HAS-RL (one-sided and two-sided) vs. unconstrained baseline at $\kappa = 5\text{ N}$ / $[\kappa_{\text{low}}, \kappa_{\text{high}}] = [3, 5]\text{ N}$ (single rollout, 100 steps).

Model	Mean force (N)	Max force (N)	Image quality	Violations
Baseline (Vanilla PPO)	5.25	5.48	0.90	96
HAS-RL (one-sided)	4.32	4.37	0.80	0
HAS-RL (two-sided)	4.04	4.09	0.77	0

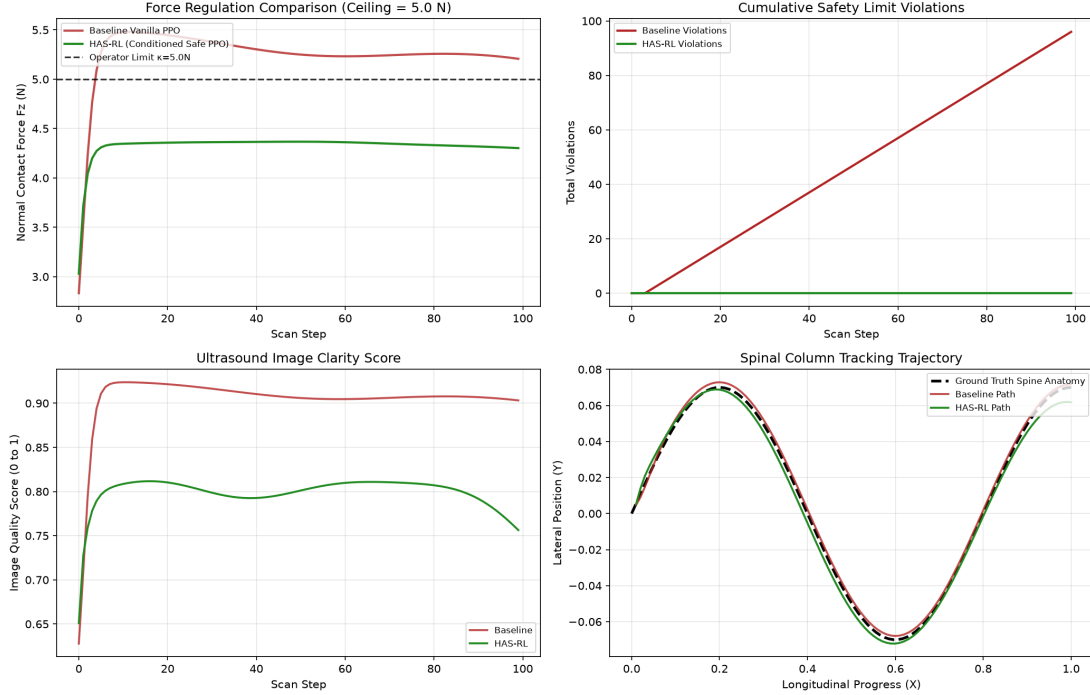


Figure 1: **HAS-RL (one-sided) competing with the baseline on one environment rollout ($\kappa = 5\text{ N}$).** Top-left: contact force vs. the operator ceiling—the baseline (red) rides above the dashed limit, HAS-RL (one-sided, green) stays below. Top-right: cumulative violations (baseline climbs to 96; HAS-RL stays at 0). Bottom-left: image-quality trace. Bottom-right: lateral tracking of the scoliotic centerline. Generated by `has_rl/interactive_demo.py`.

7.2 Operator-conditioned zero-shot safety (one-sided)

This is the headline experiment for the single-ceiling formulation. We take the *single* HAS-RL (one-sided) policy trained over the $\kappa \in [3, 12]$ curriculum and, without any further training, evaluate it at seven distinct ceilings (Table 2). Three facts stand out: (i) the learned contact force *rises monotonically* with κ —from 1.13 N at $\kappa = 3$ to 7.97 N at $\kappa = 12$, a span of +6.84 N—so the policy genuinely reads and obeys the operator dial; (ii) there are **zero violations** at every ceiling and *no* rollout ever exceeds its active ceiling; and (iii) image quality tracks the achievable acoustic window, peaking near $\kappa \approx 10$ (where the safe force approaches $F_{\text{opt}} = 7\text{ N}$) and dipping slightly at $\kappa = 12$ as the target moves past the optimum. Figure 3(a) plots force and quality against κ ; Fig. 3(b) shows the policy responding to an operator turning the dial *mid-scan* ($9 \rightarrow 4 \rightarrow 11\text{ N}$), the force following $\kappa(t)$ in real time and never crossing it.

7.3 Ablation A: anticipatory look-ahead

We retrain with the look-ahead entries of (13) enabled vs. zeroed, three seeds each, and measure lateral tracking on a deterministic rollout (Table 3). Look-ahead reduces RMS lateral error by

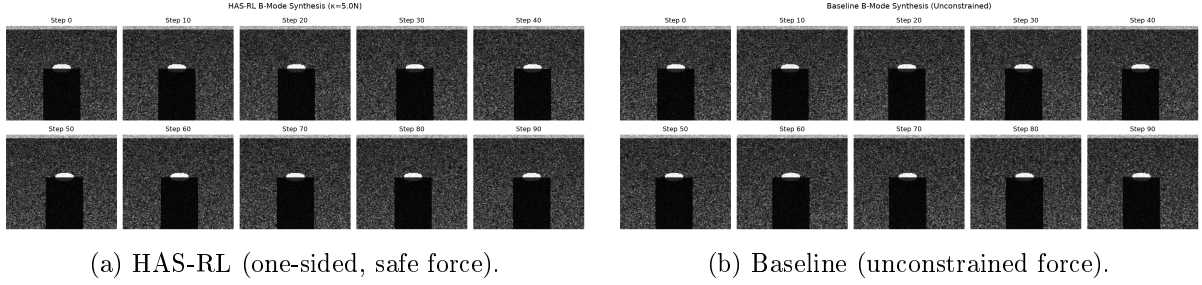


Figure 2: **Synthetic B-mode frames from the simulator** (every 10th scan step). The model renders Rayleigh speckle, depth attenuation, a hyperechoic bone arc that shifts with lateral error, and an acoustic shadow beneath the bone. Higher (unsafe) force yields brighter coupling—the very incentive the safety constraint must counter.

Table 2: Zero-shot sweep across operator ceilings for HAS-RL (one-sided), single trained policy, no retraining.

κ (N)	Mean force (N)	Max force (N)	Violations	Ceiling breach	Image quality
3	1.13	1.67	0	no	0.11
4	2.50	2.66	0	no	0.26
5	3.61	3.78	0	no	0.48
6	4.53	4.70	0	no	0.58
8	5.98	6.26	0	no	0.73
10	7.13	7.47	0	no	0.76
12	7.97	8.45	0	no	0.66
<i>Force span $\kappa=12$ minus $\kappa=3$: +6.84 N / Total violations: 0 / Breaches: 0</i>					

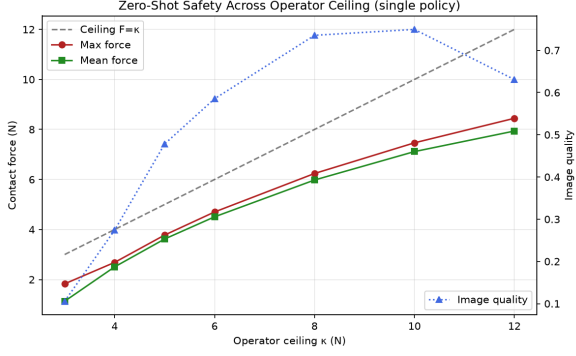
28% ($0.084 \rightarrow 0.061$) and max error by 28% ($0.137 \rightarrow 0.099$). We had also intended to report a phase-lag metric via cross-correlation, but its cross-seed standard deviation exceeded its mean (-16.7 ± 24.3 vs. -6.3 ± 28.5 steps), i.e. it is dominated by noise at this budget; we therefore report only the (clean) error reduction and do not claim a phase-lag effect.

7.4 Ablation B: multiplier-update mode (SBL vs. dual vs. raw bandit)

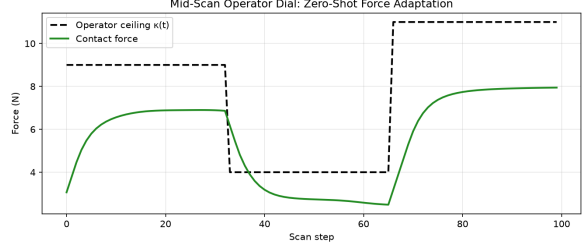
We flip only the multiplier rule (Sec. 5.2), three seeds each (Table 4). Four observations:

1. **All three are safe.** Every mode reaches *zero* evaluation violations. Part of this is by design—the budget-aware reward term (11) discourages riding the ceiling—so the multiplier is not the sole guarantor of safety here.
2. **SBL gives the best image quality** (0.603), edging out dual ascent (0.576) and clearly beating the raw bandit (0.492).
3. **The raw bandit is unstable.** Its multiplier standard deviation is 1.43 ± 0.50 —roughly $3\times$ that of SBL (0.42 ± 0.02) and dual (0.29 ± 0.10)—and this instability coincides with its lowest quality. SBL’s multiplier is not only smooth but *reproducible* across seeds (std of ± 0.02).
4. **SBL trains most stably** on the safety signal: its per-epoch violation standard deviation (11.99) is the lowest of the three (dual 16.66, raw 13.51).

Figure 4 visualizes points 3–4: the raw bandit’s λ jumps between arms epoch to epoch, dual ascent ramps slowly, and SBL rises smoothly.



(a) Force and quality vs. ceiling κ .



(b) Mid-scan dial change $9 \rightarrow 4 \rightarrow 11$ N.

Figure 3: **Zero-shot safety across the operator ceiling for HAS-RL (one-sided).** A single policy adapts its contact force to κ with no retraining and never breaches the (possibly time-varying) ceiling. Generated by `has_rl/zero_shot_eval.py`.

Table 3: Ablation A—look-ahead state augmentation (mean $\pm$ std over 3 seeds).

Look-ahead	RMS lateral error	Max lateral error
On (ours)	0.061 $\pm$ 0.060	0.099 $\pm$ 0.092
Off	0.084 $\pm$ 0.048	0.137 $\pm$ 0.068

7.5 Two-sided windows: closing the headroom gap

We train a single **HAS-RL (two-sided)** policy over the window curriculum (Sec. 5.4) and evaluate it across six operator windows (Table 5, Fig. 5). Three findings:

1. **The window holds.** Every window yields *zero* out-of-window violations and 100% in-window residence, with the force floor respected ($\min F \geq \kappa_{\text{low}}$) throughout.
2. **The gap closes.** Mean force now tracks the window, rising from 3.26 N for $[2.5, 4]$ to 10.22 N for $[9, 12]$. At the top window the two-sided policy uses 10.2 N where the one-sided ceiling model left the force at 7.97 N (Table 2)—it reclaims the ~ 3.5 N of previously wasted headroom.
3. **Quality follows physics.** Image quality peaks (0.99) for windows that bracket the acoustic optimum $F_{\text{opt}} = 7$ N (e.g. $[5, 8]$) and falls for windows forced well below or above it—an effect the one-sided model, pinned near F_{opt} , cannot exhibit.

Crucially, the two-sided constraint is materially *harder*: the active multiplier rises to $\lambda \approx 2.25$ during training versus $\lambda \approx 0.5$ in the one-sided case, i.e. the constraint is now genuinely binding rather than shadowed by the reward. This is the “harder regime” called for in Sec. 8 and a natural setting in which to re-examine the multiplier-mode comparison of Sec. 7.4.

7.6 Multiplier mode in the binding regime

Because the two-sided window makes the constraint bind ($\lambda \approx 2.25$), we repeat the multiplier-mode ablation of Sec. 7.4 with HAS-RL (two-sided) training (three seeds; Table 6). The picture is the same but sharper. All modes remain perfectly safe (zero out-of-window violations). SBL again gives the best image quality (0.715), now ahead of dual (0.677) by a wider margin than in the one-sided regime (0.038 vs. 0.027) and well clear of the raw bandit (0.594). The most striking change is *reproducibility*: SBL’s quality varies by only ± 0.011 across seeds versus ± 0.086

Table 4: Ablation B—multiplier-update mode (mean $\pm$ std over 3 seeds). Lower std = more stable. All modes achieve zero evaluation violations.

Mode	Eval violations	Image quality	train_viol_std	lambda_std
dual (PPO-Lagrangian)	0.0 $\pm$ 0.0	0.576 $\pm$ 0.138	16.66 $\pm$ 10.24	0.29 $\pm$ 0.10
exp3_raw (online bandit)	0.0 $\pm$ 0.0	0.492 $\pm$ 0.142	13.51 $\pm$ 12.76	1.43 $\pm$ 0.50
sbl (ours)	0.0 $\pm$ 0.0	0.603 $\pm$ 0.150	11.99 $\pm$ 12.44	0.42 $\pm$ 0.02

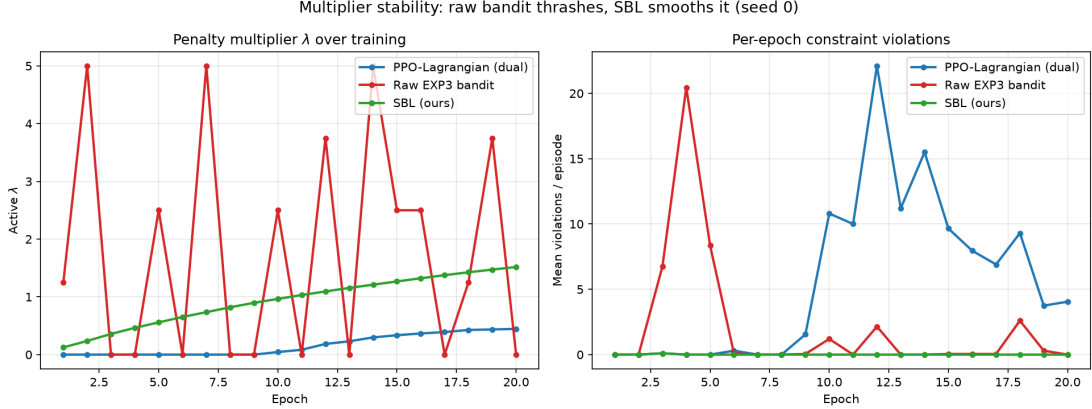


Figure 4: **Multiplier stability (seed 0).** Left: the active multiplier λ over training—the raw EXP3 bandit (red) thrashes between discrete arms, dual ascent (blue) ramps slowly, and SBL (green) evolves smoothly. Right: per-epoch violations. Generated by `has_rl/make_paper_figs.py`.

for dual—an $\sim 8\times$ tighter spread—while the raw bandit’s multiplier still thrashes ($\lambda_{\text{std}} = 1.53$ vs. SBL’s 0.45). Per-epoch violation variance is now comparable across modes (~ 21 – 23): in the binding regime the distinguishing axis is quality and its reproducibility rather than training-violation noise. We still stop short of a decisive *mean* win over dual—the 0.038 gap sits inside dual’s own cross-seed spread—but SBL is the only method that is simultaneously the highest quality, the most reproducible, and free of the raw bandit’s instability.

8 Discussion

Why Safe RL over mechanical clipping? A natural question when designing safety systems is why one cannot simply hard-code a maximum force clip on the robot’s action output rather than employing formal Safe RL machinery. If a standard RL agent is placed inside a hard-coded simulation boundary, its reward-maximizing imperative will cause it to continually command forces far exceeding the limit in pursuit of acoustic coupling. On physical hardware, constantly issuing force commands that fight a low-level safety block causes severe actuator saturation, jerky movements, and protective emergency stops. Safe RL trains the agent’s *policy* to conceptually understand the constraint. The resulting agent intelligently smooths its trajectory and holds the probe precisely at the allowed limit without fighting the system. Furthermore, learning the constraint natively is what unlocks zero-shot dynamic limits (Sec. 5.1), whereas a hard-coded limit would require halting the robot to reprogram the boundary.

What is genuinely new. The strongest result is operator-conditioned zero-shot safety: a single policy that reads a runtime ceiling and adapts its force accordingly, verified to never breach even under a mid-scan dial change (Sec. 7.2). This directly addresses the retraining

Table 5: HAS-RL (two-sided) across operator windows, single trained policy, no retraining. All windows: zero out-of-window violations, 100% in-window residence.

Window $[\kappa_{\text{low}}, \kappa_{\text{high}}]$	Mean force	Min force	Max force	In-window	Image quality
[2.5, 4]	3.26	3.23	3.33	100%	0.67
[3, 5]	4.04	4.01	4.08	100%	0.77
[4, 6]	5.02	4.99	5.09	100%	0.88
[5, 8]	6.50	6.41	7.17	100%	0.99
[7, 10]	8.36	8.35	8.38	100%	0.94
[9, 12]	10.22	9.90	10.26	100%	0.74

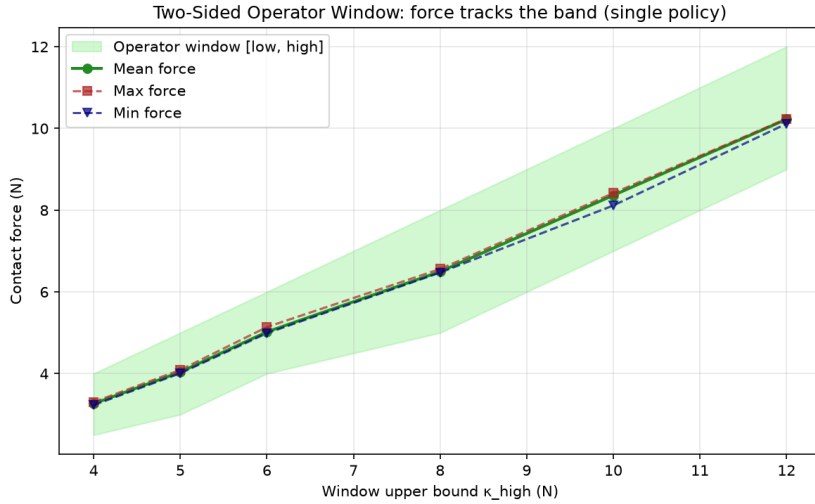


Figure 5: **HAS-RL (two-sided) operator window (single policy)**. The shaded band is the operator-specified window $[\kappa_{\text{low}}, \kappa_{\text{high}}]$; the contact force (green) rises with the window and stays inside it (min/max in blue/red), tracking the range a one-sided ceiling left unused. Generated by `has_rl/band_eval.py`.

bottleneck of fixed-threshold domain methods [2] and the context-free, offline nature of bandit-multiplier methods [3].

On SBL, honestly. SBL is favorable but its margin is modest. It achieves the best image quality and the most stable, most reproducible multiplier trajectory, at zero violations—so as a *drop-in* it is never worse than the alternatives and is the most stable of them. However, its image-quality advantage over standard dual ascent (0.603 vs. 0.576) is within one seed standard deviation, so we do not claim a statistically decisive win over dual on this reward design. SBL’s *clean* contribution is that it makes an online bandit multiplier usable at all: the raw bandit is markedly unstable (Table 4, Fig. 4), and expectation+EMA smoothing recovers dual-level performance while retaining the bandit’s bounded, interpretable penalty range. In spirit this parallels PID-Lagrangian smoothing [8]; the mechanism (bandit expectation + EMA vs. PID on a continuous multiplier) is the distinction. In the binding two-sided regime this margin widens and, more tellingly, SBL’s quality becomes far more reproducible across seeds (± 0.011 vs. dual’s ± 0.086 ; Sec. 7.6).

Why safety looks “easy” here—and how the window fixes it. Because the reward’s budget-aware term pulls force toward $\kappa - 0.4$, the one-sided reward is partly safety-aware and all multiplier modes reach zero violations, which limits how hard the constraint mechanism is

Table 6: Multiplier mode in the *binding* HAS-RL (two-sided) regime (mean $\pm$ std over 3 seeds). Compare to the one-sided Table 4: SBL’s quality lead and reproducibility both improve. All modes: zero out-of-window violations.

Mode	Eval violations	Image quality	train_viol_std	lambda_std
dual (PPO-Lagrangian)	0.0 ± 0.0	0.677 ± 0.086	21.07 ± 6.49	0.26 ± 0.09
exp3_raw (online bandit)	0.0 ± 0.0	0.594 ± 0.018	21.76 ± 7.23	1.53 ± 0.15
sbl (ours)	0.0 ± 0.0	0.715 ± 0.011	22.90 ± 7.01	0.45 ± 0.00

exercised. The two-sided operator window (Sec. 7.5) is exactly the stress test that removes this shadow: because the acoustic optimum lies outside many windows, the reward actively pushes force toward a constraint boundary, and the multiplier must climb ($\lambda \approx 2.25$ vs. ≈ 0.5) to hold the window. We ran this experiment (Sec. 7.6): in the binding regime the SBL-vs-dual quality gap widens (0.038 vs. 0.027) and SBL’s cross-seed reproducibility improves markedly, though a decisive mean win over dual remains out of reach at three seeds.

9 Limitations

Results are in a synthetic 1D-tracking simulator, not on physical hardware, real US images, or high-fidelity 3D surgical engines (e.g., SonoGym [13]); the image-quality and force models are proxies, not calibrated tissue mechanics. Ablations use three seeds—enough to expose the raw-bandit instability and the look-ahead error reduction, but too few for tight confidence intervals on small quality gaps. The spine geometry is a fixed sinusoid; anatomical variability, out-of-plane motion, and sensor noise are not modeled. The look-ahead assumes access to a short preview of the centerline, which in practice must come from a pre-scan or atlas registration.

10 Conclusion

HAS-RL turns the safety threshold of a robotic ultrasound policy from a fixed training constant into a runtime control: one policy spans a continuum of operator ceilings and adapts its contact force zero-shot, with zero violations and zero breaches across a wide sweep and under a live dial change. Alongside this we contribute SBL, a smoothed bandit-Lagrangian that makes an online multiplier bandit stable and gives the best quality at equal safety in our study, and an anticipatory look-ahead that cuts tracking error by 28%. The open CPU-only simulator makes every number here reproducible. The clear next step is a harder reward regime and a hardware or real-image validation.

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